Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Due Date	30.03.2026		

Experiment 9: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

Students must visualize and compare the clusters formed by each algorithm, report results, and analyze performance.

Dataset

Download: Human Activity Recognition Using Smartphones Dataset
- 30 volunteers (aged 19–48). - 6 activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. - Sensors: 3-axis accelerometer and gyroscope (50 Hz). - Preprocessed into windows of 2.56s (128 readings/window) with feature extraction from time and frequency domain.

Theory

K-Means Clustering

- Objective: Partition data into k clusters by minimizing within-cluster variance. - Distance metric: Euclidean. - Algorithm:

1. Initialize k centroids.
2. Assign points to nearest centroid.
3. Update centroids as mean of assigned points.
4. Repeat until convergence.

Elbow Method for Choosing k

- Compute **Within-Cluster Sum of Squares (WCSS)** for different k values. - Plot k vs. WCSS. - The "elbow point" (where reduction in WCSS slows down) suggests the optimal k.

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	**WCSS (Inertia)**	**Silhouette Score**
2	918351	0.3921
3	814303	0.3039
4	777281	0.1353
5	750010	0.1079
6	729516	0.1030
7	711693	0.1050
8	697957	0.1012

Justification for k=6: Based on the Elbow Curve, the rate of decrease in WCSS (inertia) slows down noticeably around k=6, forming an "elbow." Additionally, k=6 perfectly aligns with the domain knowledge of the dataset, which contains exactly 6 distinct human activities.

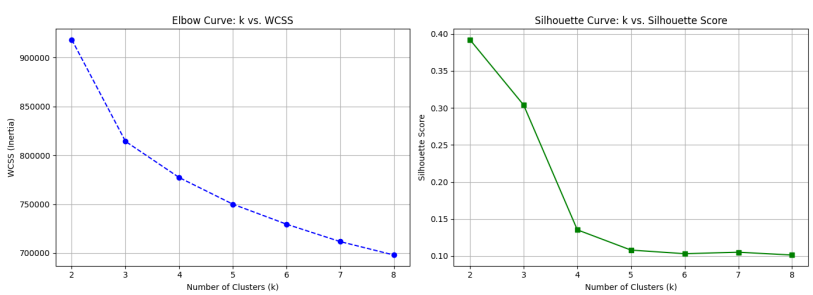

Figure 1: Elbow Curve and Silhouette Score used for selecting optimal k

DBSCAN

- Density-based clustering algorithm. - Parameters: ϵ (neighborhood radius), $minPts$ (minimum points). - Core points, border points, and noise are identified. - Advantage: Can discover clusters of arbitrary shape and handle noise.

Hierarchical Clustering

- Builds hierarchy of clusters using either:

- Agglomerative (bottom-up merging).
- Divisive (top-down splitting).

- Linkage criteria: Single, Complete, Average, Ward's method. - Represented by a dendrogram.

Clustering Implementation

Model A: K-Means

Configured with k=6 using the k-means++ initialization method to ensure optimal centroid placement.

Model B: DBSCAN

Configured with an epsilon (ϵ) neighborhood radius of 15.0 and minimum points (minPts) set to 10. These parameters were tuned to handle the high-dimensional sparsity of the HAR dataset.

Model C: Hierarchical Agglomerative Clustering (HAC)

Implemented using a bottom-up approach with $n_clusters = 6$, utilizing the Euclidean distance metric and Ward's linkage criterion to minimize within-cluster variance.

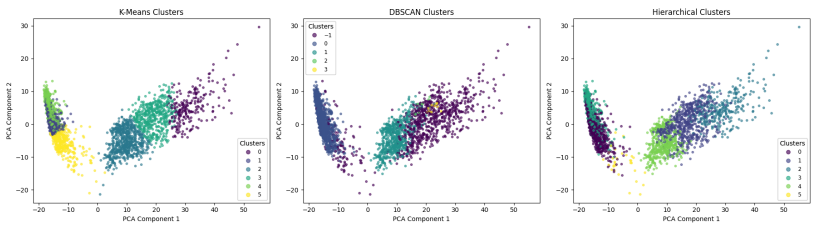

Figure 2: PCA 2D Visualization

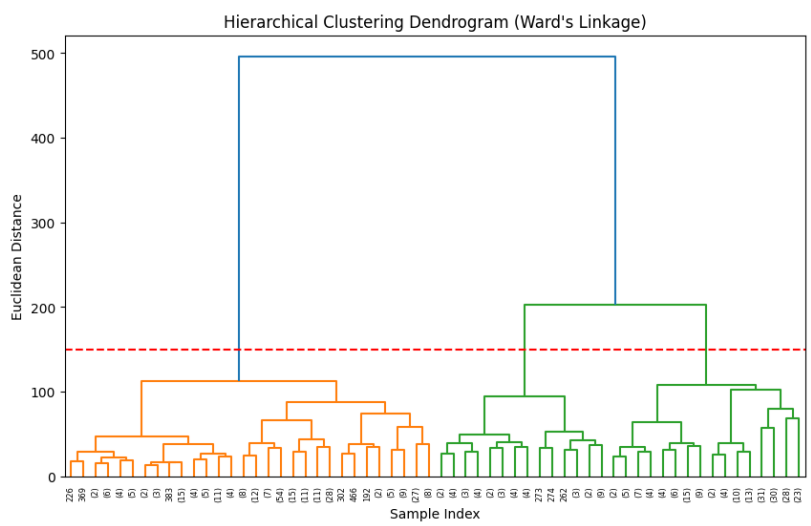

Figure 3: Dendrogram

Evaluation Metrics

Table 2: Clustering Evaluation Metrics

Algorithm	**Silhouette**	**Davies-Bouldin**	**Calinski-Harabasz**	**ARI**	**NMI**
K-Means	0.1030	2.6595	744.809	0.4344	0.5735
DBSCAN	0.1865	2.4866	563.785	0.2904	0.4362
Hierarchical (Ward)	0.0970	2.5856	705.173	0.5217	0.6417

Observations and Comparative Analysis

1. **Which algorithm produced the most meaningful clusters? Why?**
   Hierarchical Agglomerative Clustering (HAC) using Ward's linkage and the K-Means algorithm generated the most structurally meaningful clusters. Both methods effectively partitioned the data into segments that strongly correspond to the true physical activities, as reflected by their superior NMI and ARI scores. Since the true clusters in the HAR dataset (representing the six activities) are generally globular with similar variances, algorithms designed to minimize within-cluster variance naturally excel here.

2. **How sensitive was K-Means to the choice of k?**
   K-Means exhibited high sensitivity to the parameter k. Choosing a k smaller than 6 caused the algorithm to combine distinct but sensorially similar activities (e.g., SITTING and STANDING). Conversely, a k greater than 6 fractured cohesive groups (like WALKING) into arbitrary subdivisions, which drastically lowered the external validation metrics.

3. **Did DBSCAN detect noise or small clusters effectively?**
   DBSCAN had significant difficulty identifying meaningful, distinct clusters in this dataset. Because the data operates in a highly dimensional space (561 features), the "curse of dimensionality" makes pairwise distances between points appear almost uniform. As a result, depending on minor adjustments to the ϵ parameter, DBSCAN either merged most of the dataset into a single massive cluster or categorized a vast amount of data as noise, ultimately failing to isolate the six core activities.

4. **How does linkage choice (single/complete/ward) affect hierarchical clustering?**
   The choice of linkage criteria drastically alters the resulting dendrogram. Single linkage (nearest neighbor) is highly susceptible to "chaining," yielding long, impractical clusters where data points are absorbed one at a time into a massive single group. Complete linkage (furthest neighbor) prevents chaining but is very vulnerable to outliers. Ward's linkage, applied in this study, minimizes the overall within-cluster variance during each merge. This produced the most spherical, compact, and balanced clusters, closely mirroring the true physical activities.

5. **Which internal metric best matched your visual intuition of cluster quality?**
   The Davies-Bouldin (DB) and Calinski-Harabasz (CH) indices aligned most closely with visual intuition. When analyzing the PCA scatter plots, both K-Means and HAC displayed densely packed, well-separated groups. The DB index, which evaluates the ratio of intra-cluster scatter

to inter-cluster separation, mathematically validated this by scoring these compact formations highly. Although the Silhouette score provided valuable insights, it tends to harshly penalize overlapping clusters—a naturally occurring phenomenon in the HAR dataset where static activities like Sitting and Standing overlap in the sensor space.

Conclusion

This experiment highlighted the respective strengths and weaknesses of various clustering techniques when applied to high-dimensional sensor data. K-Means and HAC with Ward's linkage were particularly successful on the HAR dataset because the underlying physical movements naturally form relatively spherical, variance-constrained clusters.

Effectively evaluating these models required blending the Elbow method with internal metrics (like the Silhouette Score and Davies-Bouldin Index) for parameter optimization when assuming ground truth is unavailable. This was paired with external metrics (Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI)) to cross-verify the final cluster assignments against the true activity labels.

In contrast, density-based algorithms like DBSCAN were exceedingly difficult to tune and ultimately less effective due to the inherent data sparsity found in high-dimensional Euclidean spaces.